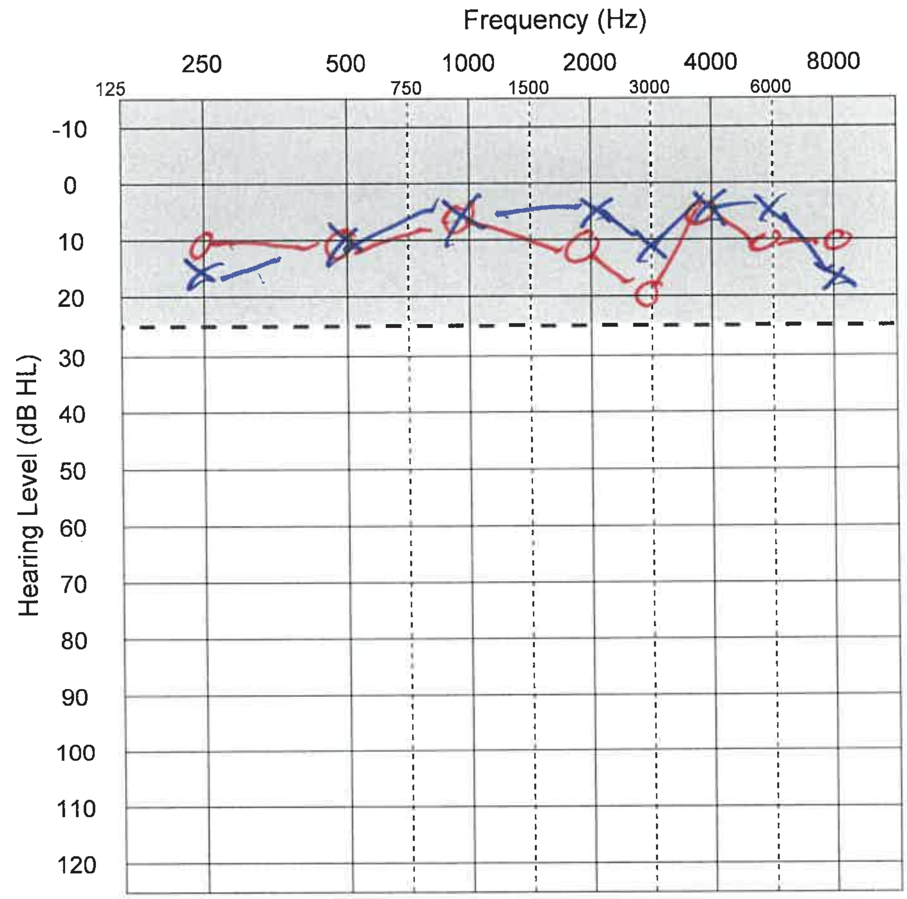
OREGON HEALTH & SCIENCE UNIVERSITY

Oregon Hearing Research Center (OHRC)
Department of Otolaryngology

Research Evaluation



Audiometer

Make/Model: GSI-61

Reliability: Good Fair Poor

PT Stimulus: Pulsed Steady FM

Transducer: Inserts TDH-50 Other:__________

Audiogram Key	Ear: Left	Ear: Unspecified	Ear: Right
Air Conduction (AC)	X		O
Masked AC	☐		Δ
Bone Conduction (BC)	>	^	<
Masked BC	]		[
Sound Field AC (Unaided)		S	
Sound Field AC (Aided)		A	
Sound Field AC (Implant)		I	
No Response	↙	↓	↘

Acoustic Immitance

	Right Ear	Left Ear
Volume (cc)	1.2	1.3
Compliance (cm^3)	0.7	0.6
Peak Pressure (daPa)	-5	-5
Gradient (daPa)	30	50

Acoustic Reflex Threshold

Probe	Stim	.5	1	2	Decay
Right	IPSI				
	Contra				
Left	IPSI				
	Contra				

Frequency (kHz)

DNT

Masking Level

		250	500	750	1000	1500	2000	3000	4000	6000	8000
RE	AC										
	BC										
LE	AC										
	BC										

Speech Audiometry

	PTA	SRT	Speech Discrimination				MCL	UCL	Lists
RE	HL	HL	%	HL	%	HL			W-22 NU-6 WIPI PB-50 PBK Other:
Masking LE									
LE	HL	HL	%	HL	%	HL			MLV Disc
Masking RE									

DNT

Remarks:

ID	RESEARCH STUDY	AGE
NH294	Other	28
SITE OF EXAMINATION	**SIGNATURE OF EXAMINING AUDIOLOGIST**	**DATE OF EXAM**
OHRC	[illegible signature]	8/8/25

FOR RESEARCH PURPOSES ONLY. FINDINGS OBTAINED IN ACCORDANCE WITH & DETERMINED BY SPECIFIC STUDY PROTOCOL. NOT INTENDED FOR HEALTH CARE MANAGEMENT.

Word Familiarity Test – Brief A

For each word below, circle the number that best reflects how familiar you are with that word.
1 = Never seen/heard   2 = Might have seen/heard   3 = Pretty sure you've seen/heard
4 = Recognize but don't know meaning   5 = Seen, vague idea of meaning
6 = Think you know meaning   7 = Confident you know meaning

1. inrush	1 2 3 4 5 6 7
2. dysprosium	1 2 3 4 5 6 7
3. chemurgic	1 2 3 4 5 6 7
4. cenobitic	1 2 3 4 5 6 7
5. obi	1 2 3 4 5 6 7
6. bosh	1 2 3 4 5 6 7
7. batrachian	1 2 3 4 5 6 7
8. bemire	1 2 3 4 5 6 7
9. appanage	1 2 3 4 5 6 7
10. citify	1 2 3 4 5 6 7
11. puttee	1 2 3 4 5 6 7
12. arable	1 2 3 4 5 6 7
13. ferrule	1 2 3 4 5 6 7
14. meliorate	1 2 3 4 5 6 7
15. capstan	1 2 3 4 5 6 7
16. hidalgo	1 2 3 4 5 6 7
17. crosier	1 2 3 4 5 6 7
18. ennui	1 2 3 4 5 6 7
19. exegesis	1 2 3 4 5 6 7
20. mullion	1 2 3 4 5 6 7
21. fief	1 2 3 4 5 6 7
22. triumvir	1 2 3 4 5 6 7
23. mastoid	1 2 3 4 5 6 7
24. hemolytic	1 2 3 4 5 6 7
25. scintillate	1 2 3 4 5 6 7
26. czarina	1 2 3 4 5 6 7
27. warder	1 2 3 4 5 6 7
28. darnel	1 2 3 4 5 6 7
29. grandiloquence	1 2 3 4 5 6 7
30. coitus	1 2 3 4 5 6 7
31. triennial	1 2 3 4 5 6 7
32. smirch	1 2 3 4 5 6 7
33. concertina	1 2 3 4 5 6 7
34. rasher	1 2 3 4 5 6 7
35. ruse	1 2 3 4 5 6 7
36. audiophile	1 2 3 4 5 6 7
37. perspicuous	1 2 3 4 5 6 7
38. stolidly	1 2 3 4 5 6 7
39. Pullman	1 2 3 4 5 6 7
40. vesture	1 2 3 4 5 6 7
41. underslung	1 2 3 4 5 6 7
42. refutation	1 2 3 4 5 6 7
43. briquette	1 2 3 4 5 6 7
44. pommel	1 2 3 4 5 6 7
45. knobbed	1 2 3 4 5 6 7
46. affray	1 2 3 4 5 6 7
47. genic	1 2 3 4 5 6 7
48. freedman	1 2 3 4 5 6 7
49. autumnal	1 2 3 4 5 6 7
50. goodwife	1 2 3 4 5 6 7
51. deluge	1 2 3 4 5 6 7
52. defy	1 2 3 4 5 6 7
53. invigorate	1 2 3 4 5 6 7
54. euphemism	1 2 3 4 5 6 7
55. dike	1 2 3 4 5 6 7
56. forked	1 2 3 4 5 6 7
57. drag	1 2 3 4 5 6 7
58. handrail	1 2 3 4 5 6 7
59. gab	1 2 3 4 5 6 7
60. destitute	1 2 3 4 5 6 7 – 134
61. antler	1 2 3 4 5 6 7
62. cannibal	1 2 3 4 5 6 7
63. quit	1 2 3 4 5 6 7
64. objectivity	1 2 3 4 5 6 7
65. comforter	1 2 3 4 5 6 7
66. greed	1 2 3 4 5 6 7
67. grab	1 2 3 4 5 6 7
68. mother	1 2 3 4 5 6 7
69. central	1 2 3 4 5 6 7
70. glory	1 2 3 4 5 6 7
71. battery	1 2 3 4 5 6 7
72. leaves	1 2 3 4 5 6 7

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NH 294

CVC-Normal Hearing (NH) Adult Data Sheet

Task	Date completed	Initials
Consent Form (circle): **Aim 2b**	8/8/25	#DVS
HIPPA		
Mini Mental State Exam (MMSE) score: 30		
Audiogram (unaided PT AC & BC, SRT & WRS), Otoscopy & Tympanometry		
Vocabulary Chart	8/28/25	

Questionnaire: **Date:** 8/28/25

Hearing History

1. Which is your dominant hand? ☒ RIGHT ☐ LEFT ☐ AMBIDEXTROUS

2. Do you have perfect pitch? ☐ YES ☒ NO

3. When was the last time you had your hearing tested? 0 8 / 2 0 2 5 (Month / Year)

4. Do you have problems hearing? ☐ YES ☒ NO ☐ UNSURE

   4a. In which ear? ☐ RIGHT only ☐ LEFT only ☐ BOTH

   4b. When did you first notice problems with your hearing? ______

5. Do you have any family members with hearing loss? ☐ YES ☒ NO

   5a. If yes, list family members, ages, and etiologies: ______

6. Do you have a history of noise exposure? ☐ YES ☒ NO

   6a. If yes, please describe type of noise exposure, length of exposure/day/week/month/year, and number of years of exposure: ______

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Age: 28

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7. Do you experience tinnitus (sounds within your ear)? ❑ YES ☒ NO ❑ UNSURE

   7a. In which ear? ❑ RIGHT only ❑ LEFT only ❑ BOTH

   7b. When did you first notice the tinnitus? |__|__| / |__|__|__|__| (Month / Year)

   7c. How would you describe the sound of your tinnitus? ____________

   7e. Are you currently experiencing tinnitus? ❑ YES ❑ NO ❑ UNSURE

   7d. How often does this occur?

   ❑ ALWAYS ❑ DAILY ☐ times/day

   ❑ WEEKLY ☐ times/week ❑ MONTHLY ☐ times/month

8. Have you previously participated in hearing-related research experiments? ☒ YES ❑ NO

   8a. If yes, describe the experiment and number of hours? ____________

Music & Language Background

9. Are you a Native English Speaker? ☒ YES ❑ NO

   9a. If yes; other spoken languages: None

   9b. If yes; speaks American English: ☒ YES ❑ NO

   9c. If yes; area you grew up in: Arizona

10. How many hours per day or week do you listen to music through headphones? 4 hrs/ day or week

11. Do you have a musical background (such as taken singing, voice, and/or instrumental music lessons, sing in a choir or play a musical instrument)? ❑ YES ☒ NO

11a. Length of your musical background: years

11b. Describe your musical background:______________________________

Audiogram

1) Copy down thresholds from audiogram into table below

Frequency	250	500	1000	1500	2000	3000	4000	6000
Left Θ	15	10	5	–	5	10	5	5
Right Θ	10	10	5	–	10	20	5	10

2) Insert audiogram as new text document into subjects folders in the C:drive and save as *Audiogram.txt*

- Date Audiogram.txt document made: 8/18/25

Vowel Introduction

❑ Go over the laminated vowel word sheet and introduce vowels (IH,EH, and AE). Explain how these are synthetic vowels and may sound a little different then use to (subject may keep laminated sheet in site during testing)

Learning Tasks

Date: 8/28/25

1) **Vocabulary Size Test**

Trial	Total Score (L):
1	218

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2) **Binaural Same Vowel CVC**
MATLAB command: CVCmono('subjectID', 'B', 66, 'y');

	.txt filename	% correct (B):
1	Experiments/Data/ NH294 \ NH294_CVCmono_0	81.82
2	Experiments/Data/ " " _1	90.91
3	Experiments/Data/ " " _2	96.97

3) **Monaural Same Vowel CVC left ear** → Subject has 3 chances to score ≥ 85%; Must do *at least 2* trials; If there is a learning effect (>5% improvement), then repeat once at the end.
MATLAB command: CVCmono('subjectID', 'L', 66, 'y');

1	Experiments/Data/ NH294 \ NH294_CVCmono_3	90.91
2	Experiments/Data/ " "	93.94
3	Experiments/Data/	
4	Experiments/Data/	

4) **Monaural Same Vowel CVC right ear** → Subject has 3 chances to score ≥ 85%; Must do *at least 2* trials
MATLAB command: CVCmono('subjectID', 'R', 66, 'y');

Trial	.txt filename:	% correct (R):
1	Experiments/Data/	93.94
2	Experiments/Data/	95.45
3	Experiments/Data/	
4	Experiments/Data/	

* Note that the fourth row is used for an additional optional repeat

*Ask the subject if they have any comments or if they heard any words/vowels that were not an option **(After first and last trial)**...*

Notes: No extra words or vowels

Monaural Qualitative Questions (Ask after general questioning at the end of the right ear trial):

1. Clarity of words: "Pretty good"
2. Vowels/Sounds heard:
3. Clarity of Each Type of Vowel: 1. IH: Good 2. EH: good 3. AE: good
4. How many vowels heard each time: 1 vowel each time

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Experimental Tasks (5 runs) **Date:** 8/28/25

1) **CVC (dichotic)** *MATLAB command*: CVC('subjectID',66,0,44);

Trial	.txt filename:	% both correct:	% >=1 correct:
1	/Experiment/Data/ *Remember to ask about participant's experience and note it in the General Notes section afterwards*	95.45 % lowlow 90.91 % highhigh 95.45 % alt Average: 93.94 %	100 % lowlow 100 %highhigh 100 % alt Average: 100 %
2	/Experiment/Data/	72.73 % lowlow 72.73 % highhigh 72.73 % alt Average: 74.24 %	95.45 ~~98.48~~ % lowlow 100 %highhigh 100 % alt Average: 98.48 %
3	/Experiment/Data/ *Remember to provide a 5-10 minute break*	68.18 % lowlow 72.73 % highhigh 63.64 % alt Average: 68.18 %	100 % lowlow 95.45 %highhigh 100 % alt Average: 98.48 %
4	/Experiment/Data/	59.09 % lowlow 63.64 % highhigh 86.36 % alt Average: 69.7 %	100 % lowlow 95.45 %highhigh 100 % alt Average: 98.48 %
5	/Experiment/Data/	59.09 % lowlow 72.73 % highhigh 63.64 % alt Average: 65.15 %	100 % lowlow 100 %highhigh 100 % alt Average: 100 %
_	/Experiment/Data/ This is an optional placeholder for redos.	____ % lowlow ____ % highhigh ____ % alt Average: ____ %	____ % lowlow ____ %highhigh ____ % alt Average: ____ %

Subject ID: NH294 Age: 28

Ask the subject if they have any comments. ***Confirm how they annotate and use the IH, EH and AE vowel buttons****. Did they hear any sounds/vowels that were not an option? Did the sound/vowel ever lateralize (ex: only heard something from the right headphone and nothing from the left)* ***(Ask after first and fifth trial)****?...*

General Notes: Heard 1 distinct "voice" and any other was unclear. "I deeply dislike that [test]" Feels unsure without the feedback

CONTINUE ON BACK⟶

Subject ID: NH294

Age: 28

Dichotic Qualitative Questions (Ask after general questioning at the end of the fifth trial):

1. Clarity of words/ Consonants: One sound was very clear. Multiple sounds not as clear
2. Vowels/Sounds heard (provide examples): 'ah'. Heard "bag" when the option was "gag" for example
3. Fusion types:
   a. Heard Three Sounds (provide examples): Center plus left right. Center was clear sound
   b. Heard Full word + Start of another word (provide examples): No
   c. Heard Full word + Only Vowel (provide examples): Deaf-ah eg
   d. Heard Differing End Consonants/ Sounds (provide examples): Not sure
4. When would you choose the word+vowel below it?: Heard word and vowel delayed
5. Lateralization (Were the different sounds always heard on different sides of the ears or was one of them heard in the center and the other heard on one side): Never just left or right
6. Did any side ever feel louder?: Left maybe slightly louder than the right
7. Did anything make it tricky?: Additional sounds were harder to get / figure out. Separating vowel from word
8. Did hearing something like a real word ~~pull~~ your answer one way?: "Does it sound like this word that rhymes"
9. What was your approach to answering these questions?: "Rambled", unsure of exact answer

❑ All tasks data backed up: Date: ________ Turn attn. back to 3 and turn off equipment. Date: ________

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